4 CV of the Experienced Researcher

AMOL DESHMUKH

ORCID ID: 0000-0003-4942-3621 ◇ Amol.Deshmukh@glasgow.ac.uk ◇ http://www.dcs.gla.ac.uk/~amol/

RESEARCH EXPERIENCE

University of Glasgow, Glasgow, UK *April 2016-current*
Research Associate MuMMer Project- European Union Horizon 2020

- MuMMer (April 2016-2019): Overall research goal of developing a humanoid robot that can interact autonomously and naturally in the dynamic environments of a public shopping mall, providing an engaging and entertaining experience to the general public.
- Developed social signal processing components to convey personality dimensions, intentions using parameterised gestures, multi-modal non-verbal behaviour generation for a shopping mall humanoid robot.
- Implemented machine learning algorithms to recognise engagement/disengagement of users, infer users' social state interest and attention, conducted user studies to evaluate their perception about the robot.

Heriot-Watt University, Edinburgh, UK *September 2008 - March 2016*
Research Associate Worked on 3 European Union FP7 projects.

- EMOTE, EU FP7, (December 2012-March 2016): Empathic robotic tutors for children in schools.
  - Evaluated interaction effects between children and a robotic tutor, such as movement, appearance, expressiveness and memory in a pedagogical setting.
  - Integrated theoretical and pedagogical work by project partners using user-centric design approach, conducted user studies with School children in Scotland.
  - Integration software modules (vision and perception), interaction modalities into a generic architectural framework.
  - Led technical meetings with project team members, produced test plans.
  - Conducted user studies with School children in Scotland.
- JAMES EU FP7 (September 2012-November 2012): A robot bartender to socially interact with humans. Developed interaction behaviours for humanoid robot, analysed video data from user studies.
- LIREC, EU FP7 (September 2008-August 2012): Interactive, emotionally intelligent companion robots.
  - Investigated the service degradation caused by recharging and explored social verbal strategies used to influence users' expectations during recharge, long-term deployment.
  - Integrated system modules, developed competencies for robot e.g. human-aware navigation, user monitoring, vision perception, autonomous recharging.
  - Designed and engineered hardware (electronics, mechanical) for mobile social robot.
  - Developed technical capabilities for social mobile robot, work-package deliverables for EU, monthly reports to EU, collaborate with other partners in the project, led technical meetings with project partners.
- Designed and constructed the mechanical and electrical components of social robot platform.
- Supervised 7 M.Sc projects, demo presentations for University open days and public events.

INDUSTRY EXPERIENCE

Bleetech Innovations, Mumbai, India *January 2018 - June 2018*
Mentor/Technical advisor

- Provided technical advise and recommend user-centred design approaches for their innovative Blee Watch, a smart watch designed for the needs of the deaf people.

Fraunhofer IPA, Neobotix, Stuttgart, Germany Feb-July 2008 & June-Aug 2007
Robotics Engineer (Internship & Masters Project)

- Thesis: "An Intelligent Human Robot Interaction System for Entertainment Mobile Robot."
- Developed a system for an entertainment mobile robot used in public places such as museums, shopping malls and exhibitions (CeBIT, Hannover, Germany).
- System Architecture: Behaviour based framework, an intelligent mechanism that accomplishes a particular task using knowledge of previous interactions with humans (using Reinforcement learning approach), ability to customise the tasks performed by the robot using a GUI.
- Monitor the performance of behaviours during interactions with humans and maintain performance scores for each behaviour, ability to customise the tasks performed by the robot.
- Implemented face detection, person tracking (OpenCV), human-Aware navigation, behaviour based control, automatic monitoring and adaptations of interactions with users.
- Mobile camera robot control programming, KUKA arm control programming, workspace designing, GUI development, Mobile platform Control (C/C++, Java).

Cosmic Technologies, Pune, India January 2006 - June 2006
Software Engineer (Internship & Master Thesis)

- Developed online Toll Free Numbers Shopping & Management System (A.S.P/ I.I.S).

Leaps & Bounds, Pune, India June 2002 - January 2003
Software Engineer: Single-handedly developed 3 web portals, full software development life-cycle.

EDUCATION

Heriot-Watt University, United Kingdom *September 2008 - December 2015*
Ph.D, "Human-Robot Interaction"

- Thesis: "An Investigation Of Service Degradation In Long-term Human-robot Interaction With a Particular Reference To Recharge Behaviour."
- Short Abstract: This work investigated service degradation in long-term interaction due to recharge behaviour of autonomous social mobile robots and proposed an approach to manage service degradation due to recharging behaviour of the robot using social verbal strategies.
- Technical Features: Human-aware navigation, path planning, vision perception, user monitoring and tracking, autonomous recharging, power management.

Jacobs University Bremen, Germany *August 2006 - August 2008*
M.Sc. Computer Science, "Smart Systems" (Scholarship Award, GPA 1.00, A+)

- Courses: Advanced robotics, Motion planning, Autonomous systems, Speech signal processing, Computer vision and perception.
- Thesis: "An Intelligent Human Robot Interaction System for Entertainment Mobile Robot".

Pune University, India *July 2003 - July 2006*
Master of Computer Application (M.C.A)

- Courses: Software Engineering, Network Programming, A.I. and Application, Database Management, Object Oriented Programming C++/ Java, Statistical Computing
- Thesis: "Online Toll Free Numbers Shopping & Management System" (A.S.P/I.I.S).

Pune University, India *July 2000 - July 2003*
Bachelor of Science, Electronics, (B.Sc.)

- Thesis: "Gyro the Humanoid robot: A self conceptualised prototype of a humanoid robot."

RESEARCH FUNDING

- Lord Kelvin Adam Smith (LKAS) Fellowship funding from UGLA £100,000, 2019.
- Travel grant of £3500 from SICSA - Scottish Informatics and Computer Science Alliance for postdoctoral and Early Career Researcher Exchange to collaborate with Amrita University in India, November 2017. (Independent research project under absence of leave).
- Contributed in defining and procuring robotics lab equipment for Heriot-Watt University, Edinburgh Centre for Robotics, 2014 (successful grant £6 million of UK government funding).

PRIZES

- Best one page statement award on "What future research is required for personal robotics and why?", Towards Autonomous Robotic Systems Conference, Ulster, 2009.
- Master's project "Gyro the humanoid robot" - 3rd Prize at Techkriti at I.I.T Kanpur (Asia's Largest Model Presentation contest), 2nd Prize at Innovations, 2nd Prize at Robotsav, got immense media publicity in India, 2005.
- Student of the year award Masters in Computer Application, 2005.

CONFERENCE ORGANISATION AND MEMBERSHIP

- Special session chair for IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN 2019), New Delhi, India, Finance chair (RO-MAN 2014).
- Main organiser for inaugural workshop on "Social robots and artificial agents for developing countries: Challenges and opportunities" (RO-MAN 2019).
- Editorial board member of new open access Journal of Future Robot Life (JFRL).
- Affiliate member of Sustainable Futures in Africa (SFA) Network, Interdisciplinary netowrk that brings together researchers, practitioners and communities to build understanding, research, and practice in socio-ecological sustainability in Africa.

RESEARCH EXPEDITIONS

- I led a team of 4 researchers from Amrita University and developed a proof-of-concept a social robot to influence handwashing behaviour in a rural village school with 100 children, 2019.
- First human-robot interaction study in a developing country to study user perception and technology acceptance of a water hauling social robot in a rural village, I led a team of 4 researchers from Amrita University, under leave of absence from UGLA, India, 2017.

DISSEMINATION ACTIVITIES

- Independently led research on human-robot interaction in rural India published in more than 15 media articles globally, March 2018. IEEE Spectrum `https://tinyurl.com/ya4mwp5l`, BBC Click `https://tinyurl.com/y53fr599`, The Telegraph `https://tinyurl.com/y5c8k3mv`, Der Spiegel.
- Speaker at Pint of science, "Robots, it makes sense!", Glasgow, 2017. `https://pintofscience.co.uk/event/robots-it-makes-sense`.
- Speaker at Nesta event, London "Humanising the robot society", June 2014, `http://www.nesta.org.uk/event/humanising-robot-society`.
- The Scottish Informatics and Computer Science Alliance (SICSA), Demofest, November 2014, URL: `https://www.youtube.com/watch?v=dUdCS1emZ5s`
- Live robot demonstration at Edinburgh International Science Festival, April 2012, "Let's Be Friends: Robot Companions".
- Video demonstration at CeBit, Hannover, Germany March 2012, "Exploring and Designing our future robot Companions".
- Featured in "Facial perception documentary", Royal Society Summer Exhibition, June 2011, `http://www.youtube.com/watch?v=Row6GSzg_m4`
- Featured on Euronews (reach 400 million homes across 160 countries) "Futuris" Program on social robotics, March 2010, `http://www.youtube.com/watch?v=qsw3tFRu8TI`

INVITED TALKS

- "Educational robots" - Invited talk at Amrita University, 2017.
- "Empathic Robotic Tutors" - Invited talk at Napier University, Edinburgh, UK 2016.
- "Gyro the humanoid robot" - Invited presentations and talks at Schools and Management colleges, India, 2005.

SELECTED PUBLICATIONS

Published 45 peer-reviewed papers (20 as first author) internationally.

1. **Amol Deshmukh** et al., "Influencing Hand-washing Behaviour With a Social Robot: HRI Study With School Children in Rural India", *The 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), October, 2019, New Delhi, India.* [accepted]
2. Unnikrishnan R, **Amol Deshmukh** et al., "Design and Perception of a Social Robot to Promote Hand Washing among Children in a Rural Indian School", *The 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), October, 2019, New Delhi, India.* [accepted]
3. **Amol Deshmukh** et al., "Shaping Robot Gestures to Shape Users' Perception: The Effect of Amplitude and Speed on Godspeed Ratings", *In 6th International Conference on Human-Agent Interaction, December, 2018, Southampton, United Kingdom. ACM*
4. **Amol Deshmukh**,et al., "Technology Acceptance, Sociocultural Influence and Gender Perception of Robots: A Human Robot Interaction Study with Naive Users in Rural India", *The 27TH IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), August, 2018, Nanjing, China.* [Independently led research]
5. **Amol Deshmukh** et al., "The More I Understand It, the Less I Like It: The Relationship between Understandability and Godspeed Scores for Robotic Gesture", *The 27TH IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), August, 2018, Nanjing, China.*
6. **Amol Deshmukh** et al., "Social Impact of recharging activity in long-term HRI and verbal strategies to manage user expectations during recharge", *Frontiers in Robotics and AI-Humanoid Robotics Journal, 2018.*
7. **Amol Deshmukh** et al., "Modulating the Non-Verbal Social Signals of a Humanoid Robot", *19th ACM International Conference on Multimodal Interaction, November 2017, Glasgow, Scotland, United Kingdom.*
8. **Amol Deshmukh**, Mary Ellen Foster and Alessandro Vinciarelli, "Perceptual Control Theory for Engagement and Disengagement of Users in Public Spaces", *International Conference on Human-Robot Interaction, Vienna, Austria, March 2017.*
9. **Amol Deshmukh** et al., "How Expressiveness of a Robotic Tutor is Perceived by Children in a Learning Environment", *International Conference on Human-Robot Interaction, Christchurch, New Zealand, March 2016.*
10. **Amol Deshmukh** et al., "An Empathic Robotic Tutor in a Map Application", *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction Extended Abstracts (Demo), 311-311, Portland, USA, March 2015.*
11. Mary Ellen Foster, **Amol Deshmukh** et al., "How does affective robot feedback influence learner experience in a real-world treasure hunt?", *In Proceedings of First International Workshop on Educational Robots (WONDER), 2015.*
12. Katrin Solveig Lohan, **Amol Deshmukh**, Ruth Aylett, "How Can a Robot Signal Its Incapability to Perform a Certain Task to Humans in an Acceptable Manner?", *In Proceedings of the 23rd IEEE International Symposium on Robot and Human Interactive Communication (IEEE RO-MAN 2014), Edinburgh, UK, 2014.*
13. **Amol Deshmukh** et al., "Towards empathic artificial tutors", *Proceedings of the 8th ACM/IEEE international conference on Human-robot interaction 2013, Tokyo, Japan.*

5 Capacity of the Participating Organisations (do i need this table at start of part b?)

Participating organisations	Legal Entity Short Name	Country	Supervisor	Role of partner organisation
Beneficiary				
University of Glasgow	UGLA	UK	Dr Mary Ellen Foster	
Partner Organisation				
Amrita Vishwa Vidyapeetham	AU	India	Prof. Bhavani Rao	Hosting the outgoing phase

Beneficiary: University of Glasgow, UGLA, UK	
General Description	Founded in 1451, the University of Glasgow has a tradition of excellence as the UK's fourth oldest university. It is a member of the elite Russell Group of leading UK research universities, a founding member of Universitas21, and of the Guild of European Research-Intensive Universities. UoG has fostered the talents of seven Nobel laureates, Scotland's first female medical graduates, and includes among its alumni, some of the world's most renowned innovators, from scientist Lord Kelvin, economist Adam Smith, to the pioneer of television John Logie Baird. Ranked 67 in the world (QS World Rankings 2020), UoG is a world-leading research-intensive institution, attracting scholars from more than 130 countries worldwide each year. In recognition of employment practices that support the careers of women in science, technology, engineering, mathematics and medicine (STEMM) in higher education, UoG has been awarded an Athena SWAN Institutional Bronze Award as well as the European Commission's 'HR excellence in research' award. The School of Computing Science (SoCS) at UGLA is in the UK's 2014 independent research exercise are rated top in Scotland for research impact with 68% of impact judged world-leading and 32% internationally excellent. In the overall research ranking, SoCS was judged equal 16th amongst UK computer science departments, rising to 10th position on research volume with 84% of all research judged world-leading or internationally excellent. SoCS are 6th in the UK on research intensity-weighted GPA rank order. SoCS is ranked 2nd in Scotland in The Complete University Guide 2019, joint 10th in the UK in THE World University Rankings by Subject 2018, and is in the top 100 in the QS World University Rankings by Subject 2018. Current research funding amounts to more than £6M in SoCS. The Social Robotics group at SoCS which is collaborative and interdisciplinary, working closely with other schools including Psychology and the Institute of Health and Wellbeing.
Academic organisation	Yes
Role and profile of key persons (supervisor)	**Dr Mary Ellen Foster** is a senior lecturer at SoCS in Human-Computer Interaction with a primary research interest in embodied communication. In particular, her expertise are in designing, implementing, and evaluating practical artificial systems that support embodied interaction with human users, such as embodied conversational agents and human-robot dialogue systems and real-world robot deployment. She is the coordinating the MuMMER project, a four-year EU Horizon 2020 project which will develop and test a socially intelligent robot in a public shopping mall. Her previous projects include COMIC (IST-2001-32311), JAST (FP6-003747-IP), and ECHOES (UK ESRC/EPSRC TEL). JAMES project (FP7-270435), which developed a socially aware robot bartender, she led the Social Signal Processing work package, and also managed the system integration and testing.
Dept.	School of Computing Science
Key research facilities, Infrastructure and Equipment	I will be provided with a fully equipped working station (with PC, printer, free access to Internet, e-mail account, telephone) with an own desk located in the main University campus. I will also have free access to UGLA's libraries; free access to various academic courses; web-profile on UGLA website; access to online scientific journals; and venues for meetings/seminars/lectures. I will have access to AU's digital resources from UGLA campus. The Human Resource Service will provide support with my contract of employment with UGLA.
Independent research premises?	Yes
Previous and current involvement in research and training programmes	UoG is currently participating in: 27 MSCA ITNs (from Horizon 2020), 42 MSCA IFs (from Horizon 2020), 5 MSCA RISE projects. The SoCS has recently been awarded a Centre for Doctoral Training (CDT) together with the Institute for Neuroscience and Psychology to train a cohort of 50 PhD students on Socially Intelligent Artificial Agents. I will have the opportunity to interact with PhD students from CDT and experts from School of Psychology.
Relevant publications and/or research/innovation products	(1) Baillie, L.; Foster, M. E.; Breazeal, C.; Fischer, K.; Denman, P.; and Cauchard, J. R The Challenges of Working on Social Robots that Collaborate with People. In CHI Conference on Human Factors in Computing Systems, (CHI'19), (2) Foster, M. E. et al., The MuMMER project: Engaging human-robot interaction in real-world public spaces. In Proceedings of the Eighth International Conference on Social Robotics (ICSR 2016), (3) Foster, M. E.; Deshmukh, A. et al., Influencing the Learning Experience Through Affective Agent Feedback in a Real-World Treasure Hunt. In Proceedings of the 14th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2015), (4) Foster, M. E. et al,. Automatically Classifying User Engagement for Dynamic Multi-party Human–Robot Interaction. International Journal of Social Robotics. July 2017, (5) Foster, M. E. Natural face-to-face conversation with socially intelligent robots. In Proceedings of the IROS 2015

Partner Organisation: Amrita Vishwa Vidyapeetham, AU, India	
General Description	Amrita Vishwa Vidyapeetham is a multi-campus, multi-disciplinary research academia that is accredited 'A' by NAAC and is ranked as one of the best research institutions in India. Amrita is spread across six campuses in three states of India - Kerala, Tamil Nadu and Karnataka, with the headquarters at Ettimadai, Coimbatore, Tamil Nadu. Amrita Vishwa Vidyapeetham continuously collaborates with top US universities including Ivy league universities and top European universities for regular student exchange programs, and has emerged as one of the fastest growing institutions of higher learning in India. The institution is managed by the Mata Amritanandamayi Math. Amrita Vishwa Vidyapeetham established under section 3 of UGC Act of 1956 offers more than 200 professional programs through 15 schools across 6 different campuses in 3 South Indian states. The world renowned humanitarian and spiritual leader Sri Mata Amritanandamayi Devi (Amma), known worldwide for Her humanitarian activities and initiatives, is the Chancellor. Amma, a great inspiration for the faculty, staff and student community founded the institution with a vision to impart high quality education enriched with human values to the younger generation for the benefit of society.Amrita Vishwa Vidyapeetham is Top Private Indian University according to QS BRICS University Rankings, a ranking of the top universities in the five BRICS countries (Brazil, Russia, India, China and South Africa). Amrita Vishwa Vidyapeetham is No. 1 in International Faculty according to QS BRICS University Rankings. Amrita Vishwa Vidyapeetham personifies multi-disciplinary academic excellence through the exceptional quality of its education system that is modern and comprehensive, with special emphasis on core human values. Amrita Vishwa Vidyapeetham was ranked 8th in India University Rankings 2019. AMMACHI Labs at AU brings an interdisciplinary approach to addressing societal challenges and as a center of Excellence, we envision a world where technology is designed and used to empower people. We create innovative educational tools and skill development solutions to help uplift rural communities. We are engaged in technology interventions and community development activities in over 142 villages across 21 states in India.
Academic organisation	Yes
Role and profile of key persons (supervisor)	**Prof. Bhavani Rao** is Director of AMMACHI Labs, she also serve as the UNESCO Chair in Gender Equality & Women's Empowerment and head the R&D activities that focuses on Human-Computer Interaction (HCI), Robotics and Automation technologies for promoting gender equality, quality STEM education, skill development, disaster response, medical assistance and rehabilitation. She is the Principal Investigator for several projects at Amrita, funded by organisations such as the United Nations Democracy Fund (UNDEF), Internet.org, NORRAG, the Ministry of Human Resources & Development (MHRD) and the National Skill Development Corporation (NSDC)
Dept.	AMMACHI Labs
Key research facilities, Infrastructure and Equipment	I will be provided with a fully equipped working station (with PC, printer, free access to Internet, e-mail account, telephone) with an own desk located in the main University campus. I will also have free access to AU's libraries; free access to various academic courses; web-profile on AU's website; access to online scientific journals; and venues for meetings/seminars/lectures. I will have access to UGLA's digital resources from AU campus.
Independent research premises?	Yes
Previous and current involvement in research and training programmes	To be added
Relevant publications and/or research/innovation products	(1) R. Unnikrishnan, Amrita, N., Muir, A., and Rao R. Bhavani, "Of elephants and nested loops: How to introduce computing to youth in rural india", in Proceedings of the The 15th International Conference on Interaction Design and Children, 2016, (2) Rao R. Bhavani, Sheshadri, S., and Unnikrishnan, R., "Vocational education technology: rural India", in Proceedings of the 1st Amrita ACM-W Celebration on Women in Computing in India, 2010 (4) Rao R. Bhavani, Rajamani, K., Bijlani, K., Achuthan, K., Sreedha, N., Nithyanandan, V., Rahul, J., and Sheshadri, S., "Virtual media enhanced vocational education curriculum", in 2010 Second International Conference on Computer Research and Development, 2010, pp. 280-284, 2018 (5) A. Radhakrishnan, Pillai, N. M., Rao R. Bhavani, Gutjahr, G., and Prof. Prema Nedungadi, "Awareness and Effectiveness of Educational Schemes for Scheduled Caste and Scheduled Tribes in Coimbatore District", International Journal of Pure and Applied Mathematics, vol. 119, pp. 1933-1941, 2018.

6 Ethical Issues

With reference to the Ethic issues table (section 4 of Part A of ENDEAVOUR), the following text is going to address the following issues: Humans, Personal data, Third countries.

ENDEAVOUR will conduct work in a manner which **complies with ethical principles**, and which adheres with existing practices carried out by AU and UGLA. Specifically, I aim to: (1) obtain verbal / written consent from each participant, making it clear that they can withdraw from the study or from answering particular questions; (2) work with school authorities; and (3) obtain verbal/written consent for video recordings.

1) Interviews, questionnaires: Prior to any interview and questionnaires, I will inform the research participants, in a language they understand (e.g. Hindi / Marathi / Malayalam), about the features and aims of ENDEAVOUR and I will explicitly ask for their informed consent. For qualitative data collection I anticipate to secure verbal informed consent, while for quantitative data collection written consent will be used (but depending on the level of literacy, verbal consent is preferred to respect participants' conditions). The verbal consent will be formally documented and independently witnessed. It will be made clear that participation is not compulsory, and participants may decline to participate in this study. The students' data including questionnaire, interview responses will anonymised no data will have information which can identify an individual, participants will be refereed with ID numbers. The same applies to semi-structured interviews and questionnaires collected from teachers on their feedback about the intervention. **No sensitive personal data** will be collected throughout the life-cycle of ENDEAVOUR.

Minors: ENDEAVOUR will involve participants who will be children aged between 5-12. I will obtain written (or verbal, depending on the level of literacy) informed consent of their parents / carers / relatives / teachers/ tutors/ legal representatives. If required / appropriate, to ensure their welfare, they will be interviewed in the presence of any of the adults above as appropriate, and, in case of girls, I will always be accompanied by a female translator. The children will be informed that if they choose, they could leave the study/interview and go back to their classrooms. ENDEAVOUR will reassure them that no coercion will ever be exerted it will be made clear to that participation is be entirely voluntary. There will not be any issue with physical harm to the participants, as the robot will be secured on a wall near water taps and the children will not be able reach the robot. There are no moving parts on the robot except for the small eyes which cannot cause any physical harm. However, there will be someone from the research team supervising the environment all the time in case they have to intervene.

2) Schools: Informed verbal/written consent will obtained from the school authorities including the principal and the teachers who will be the guardians of the children, but also the children's parents. Informed consent will obtained from the children. All data will be collected within the confines of the school. Locations for the interviews will intentionally chosen to be outside classroom and official environments. Prior to starting the research meetings will be arranged with the school authorities to introduce the project, explain the purpose of the research, and seek permission to work in the school. The reasons for conducting the work will be explained in local language. Researchers members from AU already have good rapport with the schools authorities due to their prior work so there should not any issue of trust.

Also during studies ENDEAVOUR will ensure that **no disruption** to normal school operation will happen during school hours. Interventions and interviews will be planned in the afternoon sessions during breaks. The students in the school have access to the government's scheme for free, nutritious lunches, which is a motivation for many underprivileged families to send their children to schools. In addition to this initiative, there is also a free breakfast initiative for tribal children. A typical school day starts at 10 A.M. with prayer and classes run till 12:30 P.M when the lunch break begins. In between they have a short break at 11:30 A.M. The lunch break lasts until 2:00 P.M and the classes proceed till 3:30 P.M. The intervention with the social robot will only take place between breaks to ensure not to disturb the normal school hours.

2) Photos and videos: In order to guarantee the privacy rights of the research participants, I will ask a further informed and written consent for the use of photos and interviews' videos in all ENDEAVOUR-related initiatives and materials will be obtained from their parents / carers / relatives / teachers/ tutors/ legal representatives. I will respect the subject's privacy requirements whenever they ask me to. For data

brought from India to UK, I will use automated software to pixelate faces (using Adobe After Effects) and remove audio on recorded footage after analysis has been performed.

The data generated or synthesised from ENDEAVOUR will comprise questionnaires in excel files, image and video data from field studies. All data will be made anonymous, conforming to privacy and ethical approval by AU in India and University of Glasgow ethics committee. will be stored securely with regular backup and appropriate security practices on an encrypted hard drive. Data generated will be compiled into password protected databases and files, which will be managed initially by me. Password protected access to the databases and files will be provided also to the research teams at AU and the University of Glasgow (UGLA). All data storage and provision will be compliant with GDPR procedures as implemented at UGLA. Data will be stored for a maximum period of five years following the project.

Although there are **no particular safety concerns** related to my work in India, normal safety precautions in terms of working in a foreign country and conducting field work will be taken. I will complete a full risk assessment using templates well developed by UGLA scientists conducting research in developing countries. Appropriate travel health insurance and required Pre-Exposure vaccinations will be procured, and when in the field, I will: work accompanied by at least one other person; ensure that others know the travel plans and itinerary; carry authorised identification; and carry a mobile phone. Originally being brought up in India I am well aware about the culture and social structure in India rural environments which should allow me to conduct my work comfortably.

(Do I need to fill this table below?)

Humans	
I confirm that templates of the informed consent forms and information sheets (in language and terms intelligible to the participants) will be kept on file.	Yes No
Animals	
I confirm that training certificates/personal licenses of the staff involved in animal experiments have been obtained and will be kept on file.	Yes No
I confirm that relevant authorisations for animal experiments (covering also the work with genetically modified animals, if applicable) have been obtained, and will be kept on file.	Yes No
Environmental protection and safety	
I confirm that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.	Yes No
I confirm that authorisations for relevant facilities (e.g. security classification of laboratory, GMO authorisation) have been obtained, and will be kept on file.	Yes No
Third country	
I confirm that the research performed outside the EU is compatible with the Union, National and International legislation and could have been legally conducted in one of the EU Member States.	Yes No
Data protection	
I confirm that a Data Protection Officer (DPO) has been appointed and the contact details of the DPO are made available to all data subjects involved in the research.	Yes No
I confirm that data intended to be processed is relevant and limited to the purposes of the research project (in accordance with the 'data minimisation' principle).	Yes No
I confirm that relevant authorisations for further processing of previously collected personal data have been obtained and will be kept on file.	Yes No
I confirm that the data used are publicly available.	Yes No

7 Letter of Commitment (TO BE ADDED)